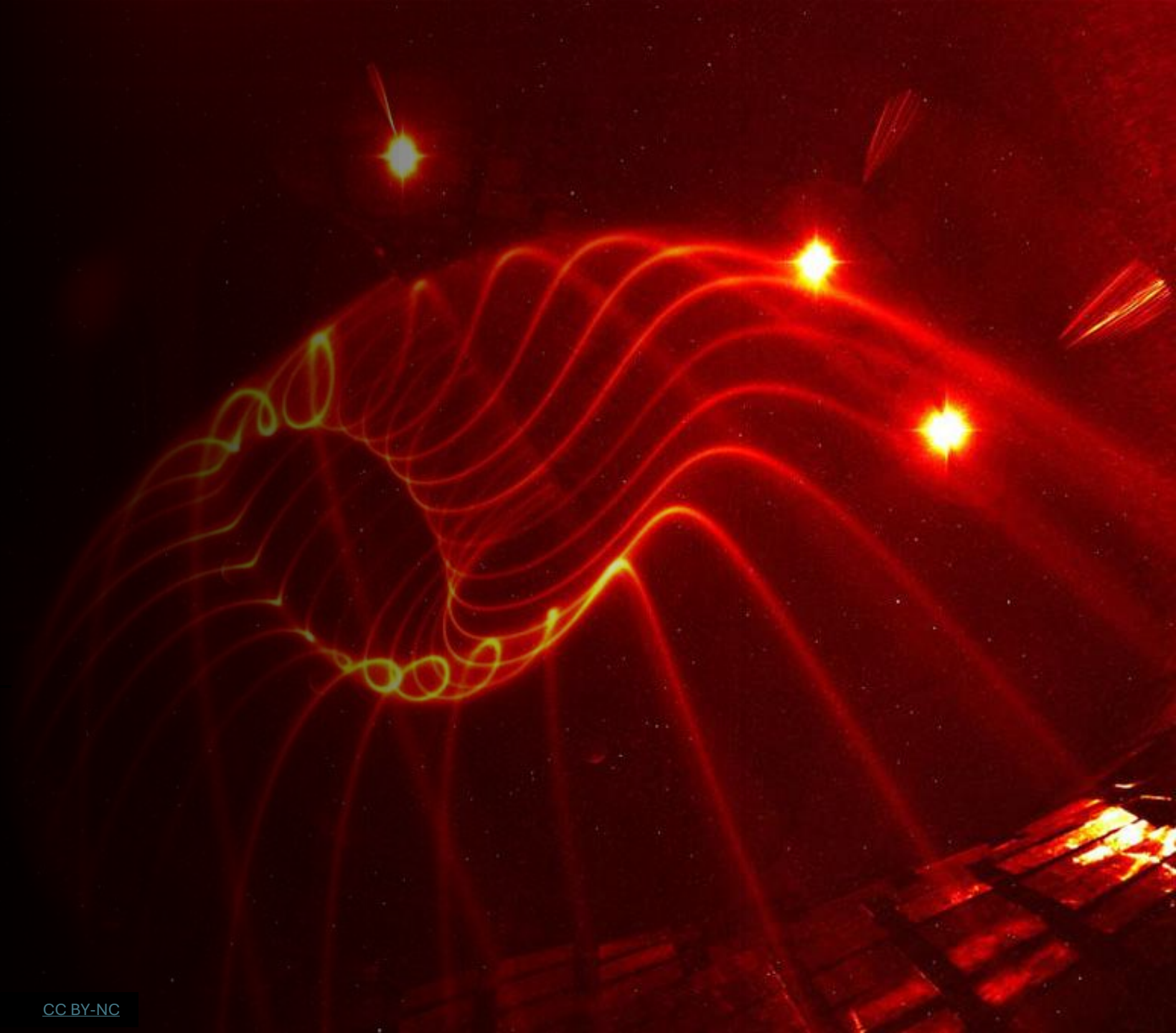




Mathematics 2 (5EZB0) Lecture 11

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Conservative fields

Conservative field

If $\mathbf{f} := (f_1, f_2, \dots, f_d) = \nabla\varphi$, for some function $\varphi: \mathcal{D} \rightarrow \mathbb{R}$, then we call $\mathbf{f}$ a conservative field and φ a scalar potential.

- They are conservative because the energy integral $\oint \mathbf{f} \cdot d\mathbf{r} = 0$ over any closed loop in $\mathcal{D}$ (to be understood in a later context)

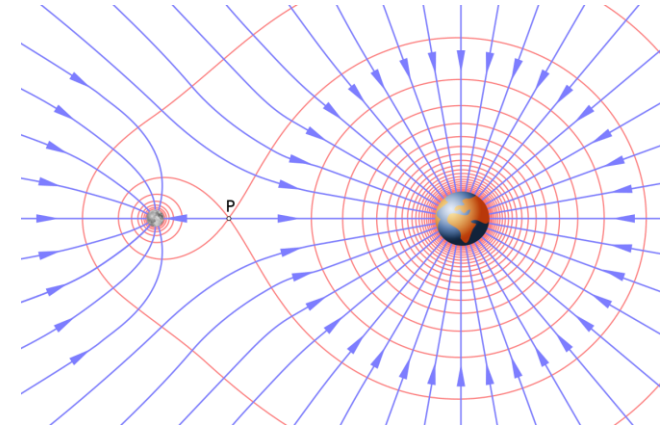
Examples

- Gravitational, electric, magnetic fields are all conservative.
- For n different masses/charges q_i at $\mathbf{x}_i$, the potential is

$$\varphi = \sum_{i=1}^n \frac{Cq_i}{|\mathbf{x} - \mathbf{x}_i|}, \quad \mathbf{f} = - \sum_{i=1}^n \frac{Cq_i(\mathbf{x} - \mathbf{x}_i)}{|\mathbf{x} - \mathbf{x}_i|^3}$$

Question 1

Show that for a conservative field $\mathbf{f}: \mathcal{D} \rightarrow \mathbb{R}$, the equipotential surfaces and the field lines are perpendicular to each other.



Equipotential surfaces

For a conservative field $\mathbf{f} = \nabla\varphi$, with potential $\varphi: \mathcal{D} \rightarrow \mathbb{R}$, the level-sets of φ are called equipotential surfaces.

Conservative fields

Conservative field

If $\mathbf{f} = \nabla\varphi$, for some function $\varphi: \mathcal{D} \rightarrow \mathbb{R}$, then we call $\mathbf{f}$ a conservative field and φ a scalar potential.

Theorem

Let $d \in \{2,3\}$, and $\mathcal{D} \subset \mathbb{R}^d$ be simply connected. Then a differentiable field $\mathbf{f}: \mathcal{D} \rightarrow \mathbb{R}^d$ is conservative if and only if for all $1 \leq i, j \leq d$,

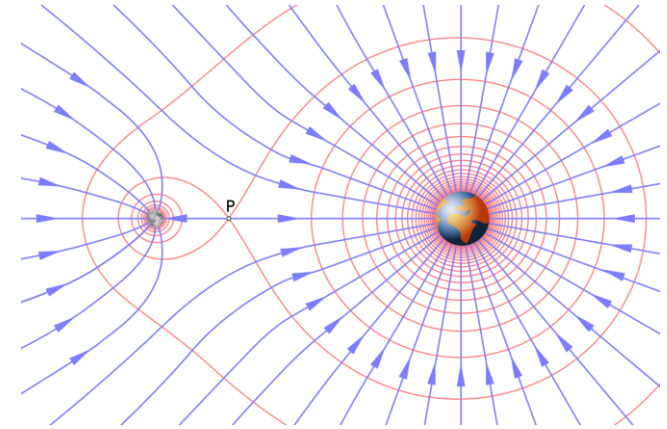
$$\frac{\partial f_i}{\partial x_j} = \frac{\partial f_j}{\partial x_i};$$

in other words, if $D\mathbf{f}$ is symmetric.

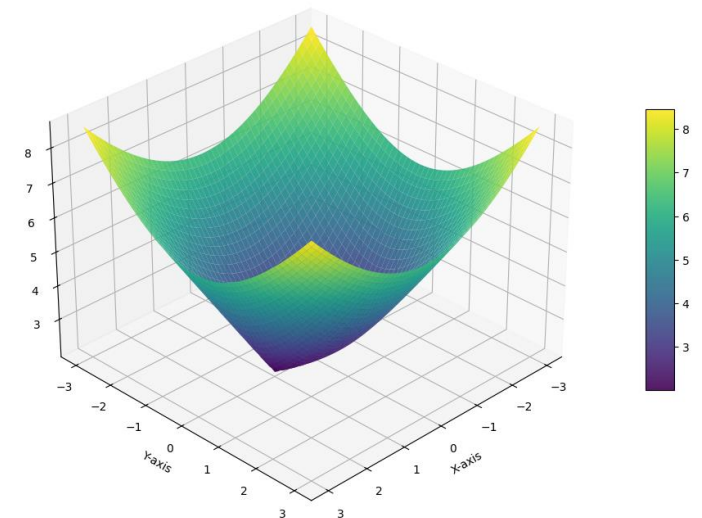
Question 2

Is the following field conservative? If so, then what is the potential and equipotential surfaces:

$$\mathbf{f}(\mathbf{x}) = \frac{\mathbf{x} - a\hat{\mathbf{i}}}{|\mathbf{x} - a\hat{\mathbf{i}}|} + \frac{\mathbf{x} + a\hat{\mathbf{i}}}{|\mathbf{x} + a\hat{\mathbf{i}}|}$$



$$z = \sqrt{(x-1)^2 + y^2} + \sqrt{(x+1)^2 + y^2}$$



Line integrals of vector fields

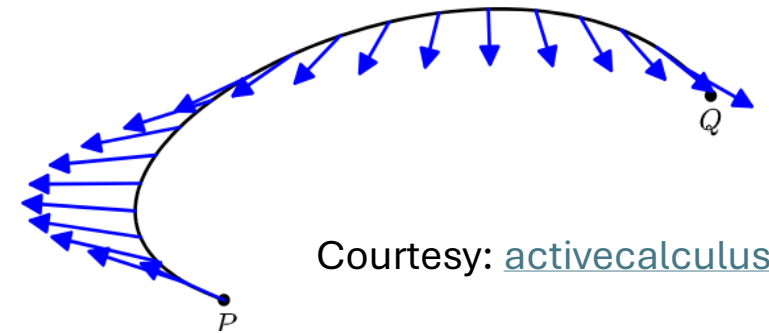
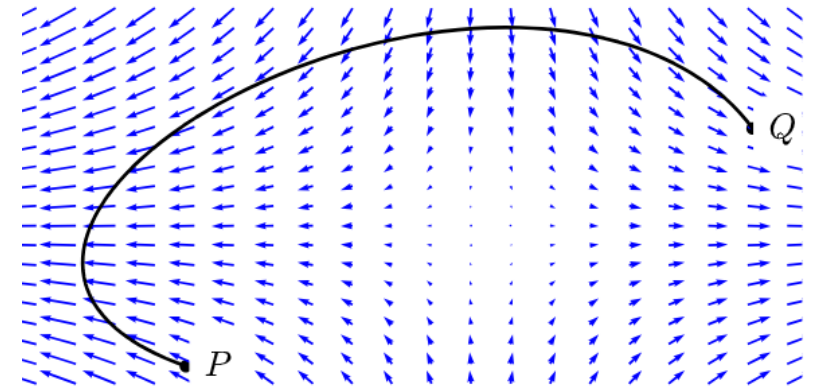
- Consider the parametric curve $\mathcal{C}$ (parameter $a \leq t \leq b$) in $\mathcal{D}$:

$$\mathbf{r}(t) = x(t) \hat{\mathbf{i}} + y(t) \hat{\mathbf{j}} + z(t) \hat{\mathbf{k}}$$
- Consider the vector field $\mathbf{f}: \mathcal{D} \rightarrow \mathbb{R}^d$
- Let $a = t_0 < t_1 < t_2 < \dots < t_n = b$ with $\Delta t = \max |t_i - t_{i-1}|$, and the Riemann sum

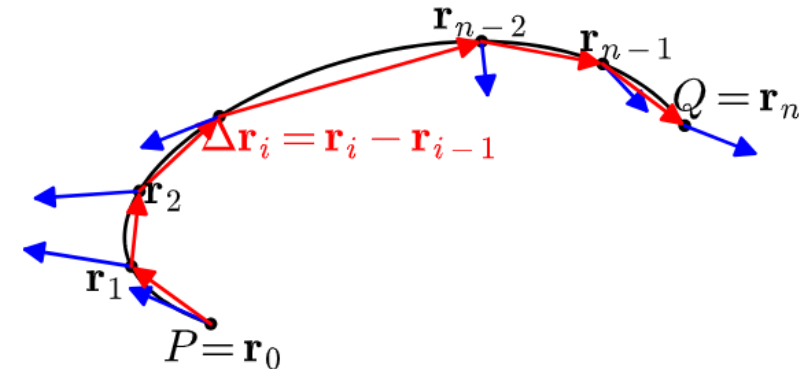
$$\sum_{i=1}^n \mathbf{f}(\mathbf{r}(t_i^*)) \cdot (\mathbf{r}(t_i) - \mathbf{r}(t_{i-1})), \quad t_i^* \in (t_{i-1}, t_i].$$

Line integral

$$\begin{aligned} \int_{\mathcal{C}} \mathbf{f}(\mathbf{r}) \cdot d\mathbf{r} &= \int_{\mathcal{C}} \sum_i^d f_i(\mathbf{r}) dx_i \\ &= \lim_{\Delta t \rightarrow 0} \sum_{i=1}^n \mathbf{f}(\mathbf{r}(t_i^*)) \cdot (\mathbf{r}(t_i) - \mathbf{r}(t_{i-1})) = \int_a^b \mathbf{f}(\mathbf{r}(t)) \cdot \frac{d\mathbf{r}}{dt} dt \end{aligned}$$



Courtesy: activecalculus.org/



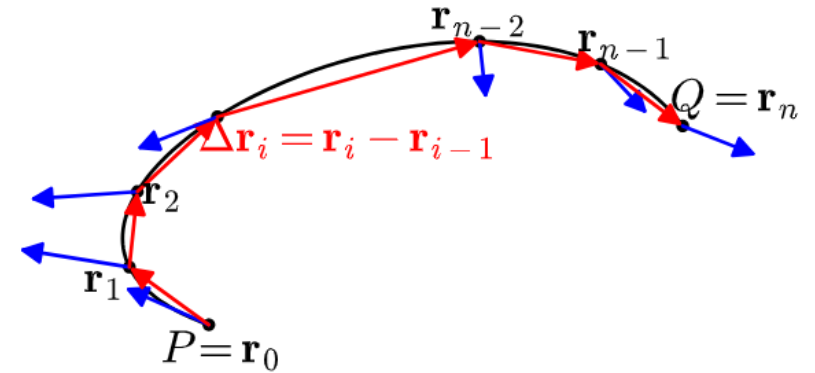
Line integrals of vector fields

- Consider the parametric curve $\mathcal{C}$ (parameter $a \leq t \leq b$) in $\mathcal{D}$:
$$\mathbf{r}(t) = x(t) \hat{\mathbf{i}} + y(t) \hat{\mathbf{j}} + z(t) \hat{\mathbf{k}}$$
- Consider the vector field $\mathbf{f}: \mathcal{D} \rightarrow \mathbb{R}^d$
- Let $a = t_0 < t_1 < t_2 < \dots < t_n = b$ with $\Delta t = \max |t_i - t_{i-1}|$, and the Riemann sum

$$\sum_{i=1}^n \mathbf{f}(\mathbf{r}(t_i^*)) \cdot (\mathbf{r}(t_i) - \mathbf{r}(t_{i-1})), \quad t_i^* \in (t_{i-1}, t_i].$$

Line integral

$$\begin{aligned} \int_{\mathcal{C}} \mathbf{f}(\mathbf{r}) \cdot d\mathbf{r} &= \int_{\mathcal{C}} \sum_i^d f_i(\mathbf{r}) dx_i \\ &= \lim_{\Delta t \rightarrow 0} \sum_{i=1}^n \mathbf{f}(\mathbf{r}(t_i^*)) \cdot (\mathbf{r}(t_i) - \mathbf{r}(t_{i-1})) = \int_a^b \mathbf{f}(\mathbf{r}(t)) \cdot \frac{d\mathbf{r}}{dt} dt \end{aligned}$$



Run **line_integral.m** to see how for an arbitrary vector field and curve the Riemann sum converges to the line integral

Line integrals of vector fields

- Consider the parametric curve $\mathcal{C}$ (parameter $a \leq t \leq b$) in $\mathcal{D}$:

$$\mathbf{r}(t) = x(t) \hat{\mathbf{i}} + y(t) \hat{\mathbf{j}} + z(t) \hat{\mathbf{k}}$$

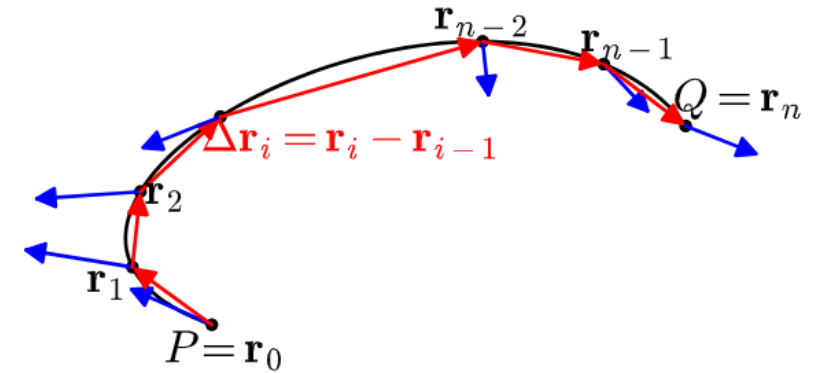
- Consider the vector field $\mathbf{f}: \mathcal{D} \rightarrow \mathbb{R}^d$

- Let $a = t_0 < t_1 < t_2 < \dots < t_n = b$ with $\Delta t = \max |t_i - t_{i-1}|$, and the Riemann sum

$$\sum_{i=1}^n \mathbf{f}(\mathbf{r}(t_i^*)) \cdot (\mathbf{r}(t_i) - \mathbf{r}(t_{i-1})), \quad t_i^* \in (t_{i-1}, t_i].$$

Line integral

$$\begin{aligned} \int_{\mathcal{C}} \mathbf{f}(\mathbf{r}) \cdot d\mathbf{r} &= \int_{\mathcal{C}} \sum_i^d f_i(\mathbf{r}) dx_i \\ &= \lim_{\Delta t \rightarrow 0} \sum_{i=1}^n \mathbf{f}(\mathbf{r}(t_i^*)) \cdot (\mathbf{r}(t_i) - \mathbf{r}(t_{i-1})) = \int_a^b \mathbf{f}(\mathbf{r}(t)) \cdot \frac{d\mathbf{r}}{dt} dt \end{aligned}$$



Notes

- Does not depend on the parametrization chosen: for reparametrization $t \mapsto s$,

$$\begin{aligned} &\int \mathbf{f}(\mathbf{r}(t)) \cdot \frac{d\mathbf{r}}{dt} dt \\ &= \int \mathbf{f}(\mathbf{r}(s)) \cdot \frac{d\mathbf{r}}{ds} \frac{ds}{dt} dt \\ &= \int \mathbf{f}(\mathbf{r}(s)) \cdot \frac{d\mathbf{r}}{ds} ds \end{aligned}$$

Line integrals of scalar fields

- Consider the parametric curve $\mathcal{C}$ (parameter $a \leq t \leq b$) in $\mathcal{D}$:

$$\mathbf{r}(t) = x(t) \hat{\mathbf{i}} + y(t) \hat{\mathbf{j}} + z(t) \hat{\mathbf{k}}$$

- Consider the scalar field $f: \mathcal{D} \rightarrow \mathbb{R}$

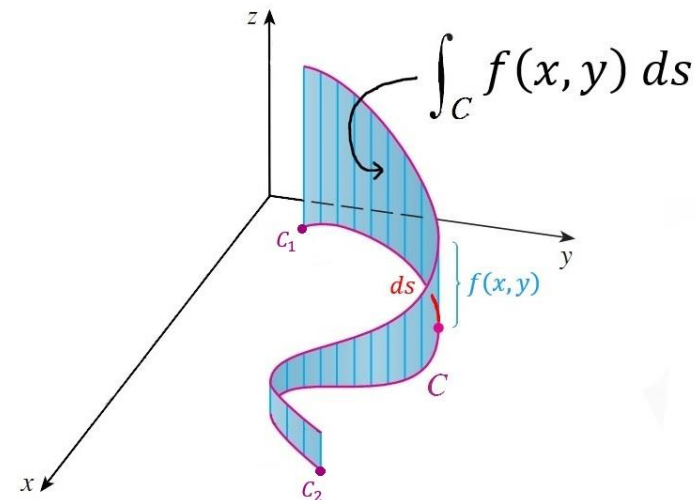
- Let $a = t_0 < t_1 < t_2 < \dots < t_n = b$ with $\Delta t = \max |t_i - t_{i-1}|$, and the Riemann sum

$$\sum_{i=1}^n f(\mathbf{r}(t_i^*)) |\mathbf{r}(t_i) - \mathbf{r}(t_{i-1})|, \quad t_i^* \in (t_{i-1}, t_i].$$

Line integral

Let s be the intrinsic (arc-length) parametrization of $\mathcal{C}$. Then

$$\begin{aligned} \int_{\mathcal{C}} f(\mathbf{r}(s)) ds &= \lim_{\Delta t \rightarrow 0} \sum_{i=1}^n f(\mathbf{r}(t_i^*)) |\mathbf{r}(t_i) - \mathbf{r}(t_{i-1})| \\ &= \int_a^b f(\mathbf{r}(t)) \left| \frac{d\mathbf{r}}{dt} \right| dt \end{aligned}$$



Notes

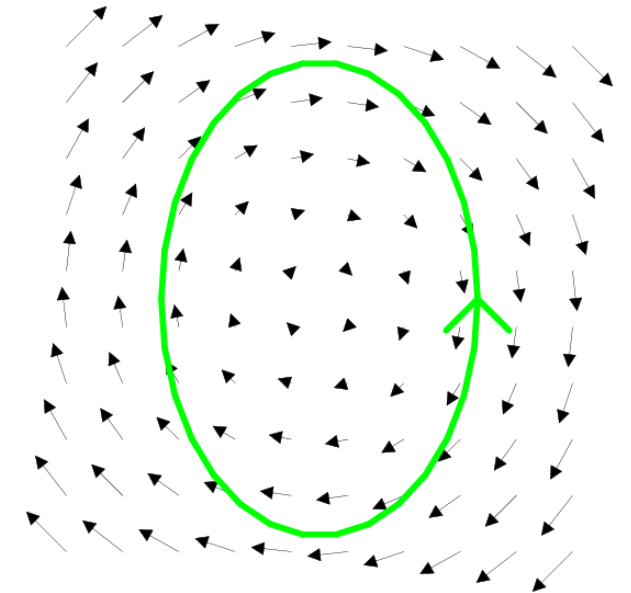
- Does not depend on parametrization as before.
- To compute the length of a curve $\mathcal{C}$ is to simply set $f = 1$, i.e., the arc-length is $\int_{\mathcal{C}} ds$

Different line integrals

Line integral

$$\int_C \mathbf{f}(\mathbf{r}) \cdot d\mathbf{r} = \int_a^b \mathbf{f}(\mathbf{r}(t)) \cdot \frac{d\mathbf{r}}{dt} dt$$

$$\int_C f(\mathbf{r}(t)) ds = \int_a^b f(\mathbf{r}(t)) \left| \frac{d\mathbf{r}}{dt} \right| dt$$



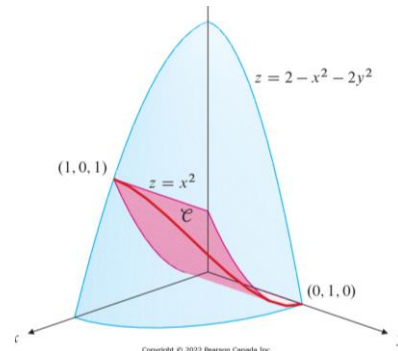
➤ If C is closed, then the line integral is called *the circulation*

$$\oint_C \mathbf{f}(\mathbf{r}) \cdot d\mathbf{r}$$

Question 3.1

Find the following line integrals

1. $\int_C z \, ds$ for the conical helix $(t \cos t, t \sin t, t)$ if $0 < t < 2\pi$, i.e., its z -centroid
2. Example 4 in Ch 16.3

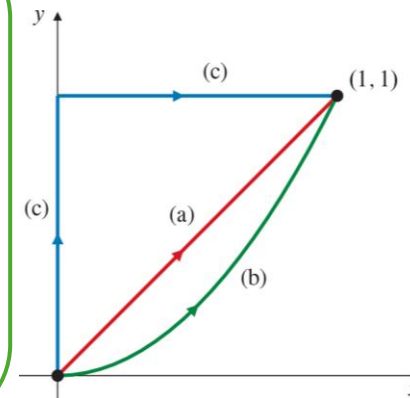


Question 3.2

Find the line integral of $\mathbf{f} = x^2\hat{\mathbf{i}} - y\hat{\mathbf{j}}$ along the following curves C connecting $(0,0)$ and $(1,1)$

1. C is composed of the line $\{x = 0, 0 \leq y \leq 1\}$ and the line $\{0 \leq x \leq 1, y = 1\}$.
2. C is the curve $\{0 \leq x \leq 1, y = x^2\}$

What do you notice?



Conservative field

Line integral of conservative fields

Let $\mathbf{f} = \nabla\varphi$ be a conservative field. Then along a curve $\mathcal{C}$ connecting $\mathbf{r}(a) = \mathbf{P}$ and $\mathbf{r}(b) = \mathbf{Q}$ and entirely belonging in $\mathcal{D}$:

$$\int_{\mathcal{C}} \mathbf{f}(\mathbf{r}) \cdot d\mathbf{r} = \varphi(\mathbf{Q}) - \varphi(\mathbf{P}),$$

Implying that

$$\oint_{\mathcal{C}} \mathbf{f}(\mathbf{r}) \cdot d\mathbf{r} = 0$$

➤ If $\mathbf{f}$ is a conservative force, then the integral is called the **work**

Question 3.1

For motion following the Newton's law,

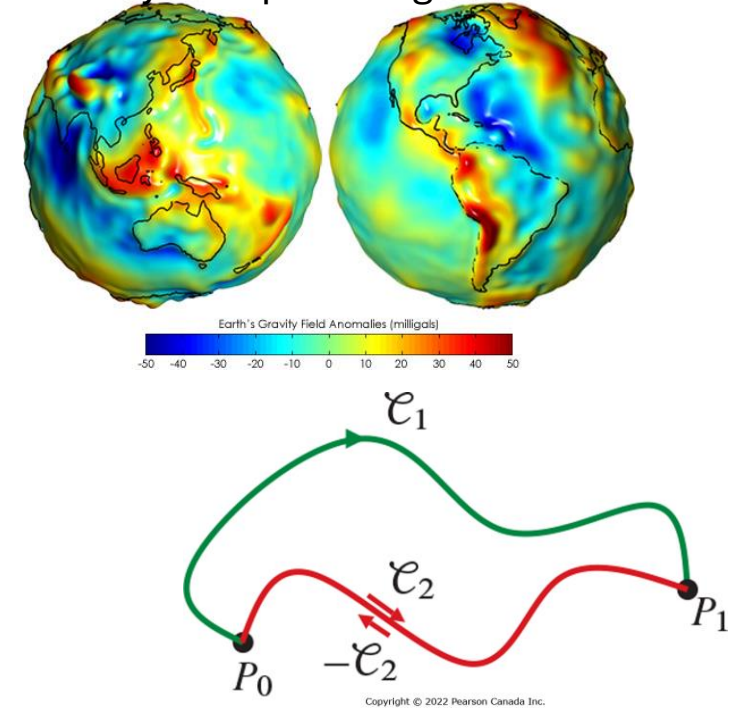
$$m \frac{d^2 \mathbf{r}}{dt^2} = \mathbf{f}(\mathbf{r}(t))$$

show the conservation of energy $\mathcal{E} = \frac{1}{2} m \left(\frac{d\mathbf{r}}{dt} \right)^2 - \varphi(\mathbf{r})$

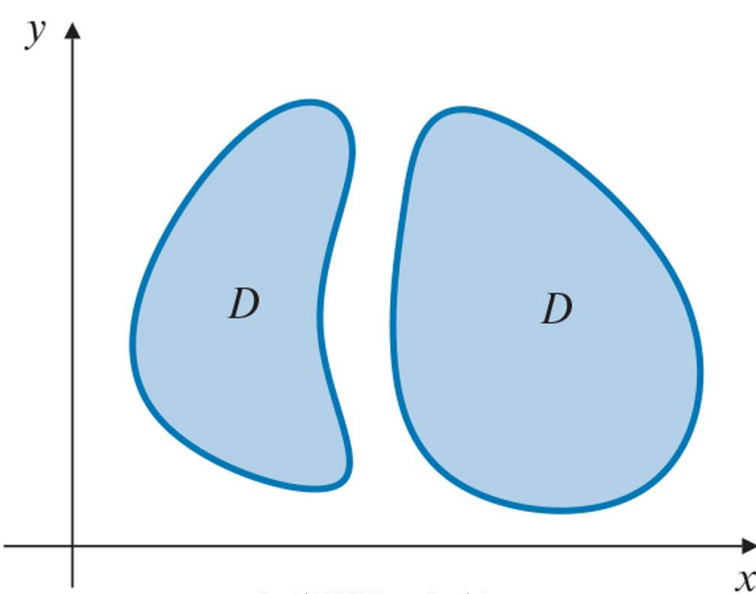
Question 3.2

Show that if $\oint_{\mathcal{C}} \mathbf{f}(\mathbf{r}) \cdot d\mathbf{r} = 0$ for all closed curves $\mathcal{C}$ for some vector field $\mathbf{f}$, then value of any line integral only depends on the end-points.

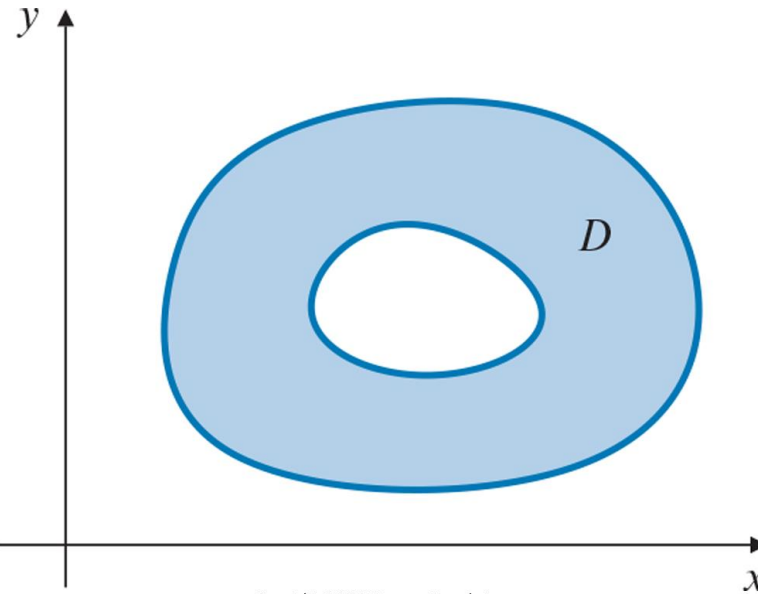
Courtesy: Wikipedia.org



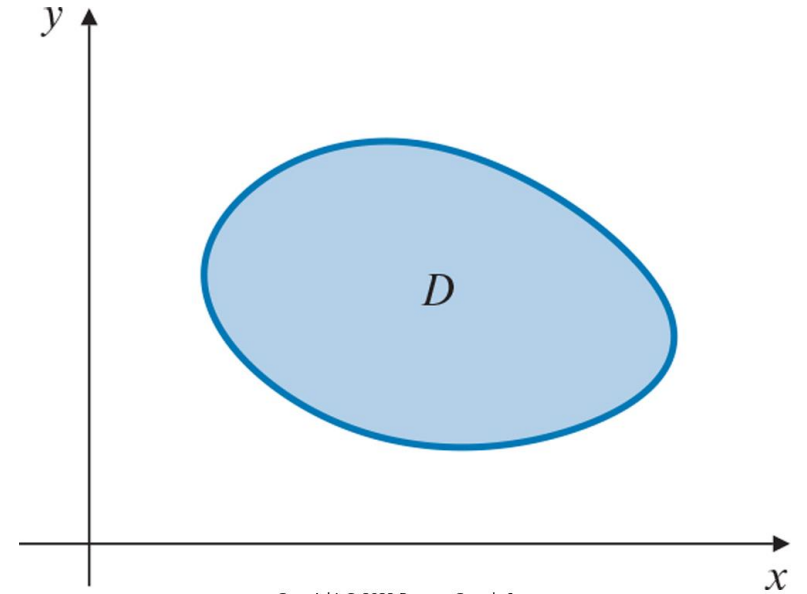
Conservative field



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disconnected domains

There exists two points which cannot be joined by a curve lying entirely in $\mathcal{D}$

Connected domains

A domain $\mathcal{D}$ is said to be connected if every pair of points $\mathbf{P}$ and $\mathbf{Q}$ can be joined by a piecewise smooth curve lying in $\mathcal{D}$.

Simply connected domains

A connected domain $\mathcal{D}$ in which every simple closed curve can be shrunk continuously to a point in $\mathcal{D}$ without ever passing out of $\mathcal{D}$.

Conservative field theorem

Conservative Field Theorem

Let $d \in \{2,3\}$, and $\mathcal{D} \subset \mathbb{R}^d$ be simply connected, and $\mathbf{f}: \mathcal{D} \rightarrow \mathbb{R}^d$ be a continuously differentiable field. Then the following statements are equivalent

- a) $\mathbf{f}$ is conservative, i.e., $\mathbf{f} = \nabla \phi$.
- b) the integral $\int_{\mathcal{C}} \mathbf{f}(\mathbf{r}) \cdot d\mathbf{r}$ only depends on the endpoints of $\mathcal{C}$
- c) The integral $\oint_{\mathcal{C}} \mathbf{f}(\mathbf{r}) \cdot d\mathbf{r} = 0$ for all closed $\mathcal{C}$
- d) for all $1 \leq i, j \leq d$,

$$\frac{\partial f_i}{\partial x_j} = \frac{\partial f_j}{\partial x_i}.$$

In other words, $D\mathbf{f}$ is symmetric, or $\nabla \times \mathbf{f} = \mathbf{0}$

Question 4

Consider the rotational field

$$\mathbf{f}(x, y) = \omega(-y \hat{\mathbf{i}} + x \hat{\mathbf{j}}) = (\omega \hat{\mathbf{k}}) \times \mathbf{r}.$$

Show that none of the conditions above are satisfied.

Line integral

$$\int_{\mathcal{C}} \mathbf{f}(\mathbf{r}) \cdot d\mathbf{r} = \int_a^b \mathbf{f}(\mathbf{r}(t)) \cdot \frac{d\mathbf{r}}{dt} dt$$
$$\int_{\mathcal{C}} f(\mathbf{r}(t)) ds = \int_a^b f(\mathbf{r}(t)) \left| \frac{d\mathbf{r}}{dt} \right| dt$$

Next class- Flux integrals

